Work-energy lab

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Compare measured work with kinetic-energy change.

SUCCESS CRITERIA - I can...

1. Find work from force-displacement area.
2. Calculate kinetic energy from measured mass and speed.
3. Discuss disagreements without inventing measurements.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “trial” and “work” to explain your answer. Sentence starters are optional.

trial: one run or repetition of a test

work: energy transferred by a force along a displacement

displacement: change from initial position to final position, with direction

01 Warm-up: bring back an earlier idea

A **3 N** friction force opposes a **5 m** displacement. **Find work**, including sign and units.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $W = F_{parallel}\, d$; $W_{net} = \Delta K$.
Constant force for the first equation; otherwise use signed force-displacement area. Use the approved cart setup and **record actual trials separately**.

Worked reasoning: Supplied measurements give net work **12 J** and kinetic-energy change **11 J**. Their **1 J** difference could reflect uncertainty or omitted work; **check this difference**.

Your turn: Draw two bars for these measurements and **label one check of the** experiment.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Initial **Transfers** **Final**

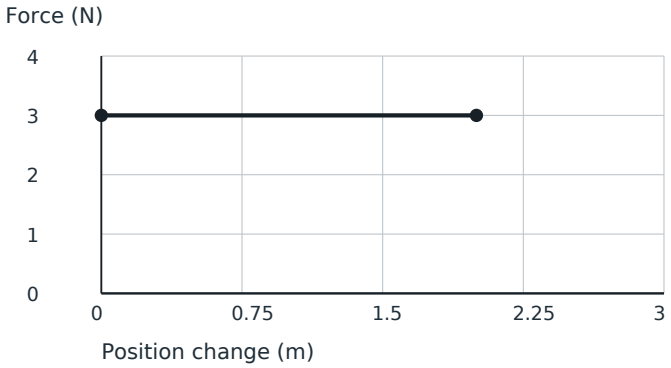
Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

The supplied force-displacement graph stays at **3 N** over **2 m**. **Calculate work**.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: ① ② ③

04 Apply the physics

A **2 kg** cart slows from **3 m/s** to **2 m/s** during a measured interval. A force sensor reports **+3 J** of net work over that same interval. **Find initial K**, final K and ΔK. Do the two measurements agree? Suggest a check.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

1. Create a trial table with mass, initial/final speed, work and ΔK.
2. **Name two causes of disagreement** that deserve checking.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Two equal **4 kg** carts approach at **6 m/s** in opposite directions and stick. **Find their common final velocity** and explain.
Use before and after snapshots to organize the momentum account.

Word help: final = at the end of the interval; subscript f

Sentence starter: I think … . I can point to …, which shows …, so … .

Before	After

Label mass, velocity and momentum.

Marzano check - circle: 1 2 3

07 Exit check

Why must the work and speed change refer to the same start and end positions? **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Sentence starter: Both … . They differ in … because … .

Initial	Transfers	Final

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Two trials have differences **1 J** and **5 J**, each with possible difference uncertainty **±2 J**. Which is consistent with zero under those bounds? **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Sentence starter: Both … . They differ in … because … .

Initial	Transfers	Final

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3